



Figure 3: Status check for a valid session

The login script obtains the IP address of the remote client using the global variable `$_SERVER[REMOTE_ADDR]`. The user ID is obtained from the user table. The starting and last access times are also inserted into the table. The script begins working from the *if* condition found at the foot, calls *dblogin()* function and waits for the boolean result to arrive from the function. The function returns true only if all the checking operations are successfully completed.

Checking session status

Clicking the 'Check session status!' link takes us to the status checking script, named *dbloginstatus.php*. This script checks whether the IP address of the machine requesting the page holds a valid session. The session might expire either by calling *dblogout.php* (clicking the link or directly entering the URL) or by time out. The time limit is specified by the *\$time_out* variable, which stores the maximum permissible time since the last access. In the present case, the time limit was set to 120 seconds, to quickly test the time-out message. But, in the case of real-world projects, it may vary from 1,200 seconds (20 minutes) to 2,400 seconds (40 minutes).

Status messages for valid sessions and timed-out sessions are shown in Figures 3 and 4. The status monitoring script, given below, should be included at the top of all the pages that require to be protected for the access of valid users:

```
<?php
/*dbloginstatus.php*/
$user;
$session_id;
$time_out=120;
function checkStatus() {
    global $time_out;
    $uid;
    global $user, $session_id;
    $valid=false;
    $remote_ip = $_SERVER[REMOTE_ADDR];
```

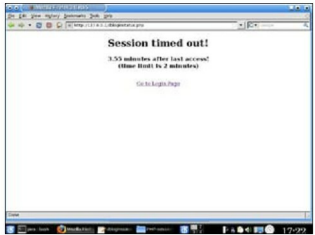


Figure 4: Status check for a timed-out session

```
$con = mysql_connect('localhost','user','pass') or die(mysql_error());
mysql_select_db('session', $con) or die(mysql_error());
$res = mysql_query("select user_id, session_id, status,
time_to_sec(timediff(now(), last_access)) as duration from session_log
where remote_ip='$remote_ip' and session_id=(select max(session_id) from
session_log where remote_ip='$remote_ip')", $con) or die(mysql_error());
if(mysql_num_rows($res) == 0) {
    print("<center><h1>Invalid Session!</h1><br />");
    mysql_close($con);
    print("<a href='\"dbloginform.html\"'>Go to Login
Page</a></center>");
    exit(0);
}

$session_id = mysql_result($res,0,"session_id");
$uid = mysql_result($res,0,"user_id");
$d = mysql_result($res,0,"duration");
$n=mysql_num_rows($res);
$st=mysql_result($res,0,"status");
if($st == "EXPIRED") {
    print("<center><h1>Invalid Session!</h1><br />");
    mysql_close($con);
    print("<a href='\"dbloginform.html\"'>Go to Login
Page</a></center>");
    exit(0);
}
else if($d > $time_out) {
    print("<center><h1>Session timed out!</h1><h3>
. ($d/60). " minutes after last access!<br />(time limit is " . ($time_out/60) . "
minutes)</h3><br />");
    $res = mysql_query("select @m:=max(session_id)
from session_log where remote_ip='$remote_ip'", $con);
    $res = mysql_query("update session_log set
status=2 where remote_ip='$remote_ip' and session_id=@m", $con) or
die(mysql_error());
    mysql_close($con);
    print("<a href='\"dbloginform.html\"'>Go to Login
Page</a></center>");
```